A GERBE WHOSE TOTAL SPACE IS SMOOTH OVER THE BASE HAS A SMOOTH STRUCTURE MORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the structure morphism of a gerbe is flat and locally of finite presentation. Invoked at the start of the counted proof (source line 7734). Supplies one half of the smoothness assertion.
- The descent statement for smoothness along a flat finitely presented cover. Invoked inside the counted proof (source lines 7735-7737). Reduces the assertion to a flat finitely presented base change.
- The local structure of a gerbe as a quotient stack over an algebraic space. Invoked inside the counted proof (source lines 7738-7741). Presents the base-changed gerbe as a quotient by a flat finitely presented group.
- The statement that a quotient of a smooth space by a flat finitely presented group is smooth over the base. Invoked at the end of the counted proof (source line 7744). Gives smoothness in the quotient case.

Source reference. Original lemma lemma-descent-smooth: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7627–7627. Navigation pointer only: the counted text refers to the Stacks item lemma-descent-smooth, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-gerbe-fppf: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6287–6287. Navigation pointer only: the counted text refers to the Stacks item lemma-gerbe-fppf, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-local-structure-gerbe: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6248–6248. Navigation pointer only: the counted text refers to the Stacks item lemma-local-structure-gerbe, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-smooth-quotient-stack: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7698–7698. Navigation pointer only: the counted text refers to the Stacks item lemma-smooth-quotient-stack, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $\pi : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. If $\mathcal{X}$ is a gerbe over $\mathcal{Y}$, then π is surjective and smooth.*

Proof. We have seen surjectivity in Lemma lemma-gerbe-fppf. By Lemma lemma-descent-smooth it suffices to prove to the lemma after replacing π by a base change with a surjective, flat, locally finitely presented morphism $\mathcal{Y}' \to \mathcal{Y}$. By Lemma lemma-local-structure-gerbe we may assume $\mathcal{Y} = U$ is an algebraic space and $\mathcal{X} = [U/G]$ over U with $G \to U$ flat and locally of finite presentation. Then we win by Lemma lemma-smooth-quotient-stack. □